



Opera Numerorum -- frontispiece. Artistic illustration generated with AI (Replit media generation). [ARTISTIC ILLUSTRATION]

Opera Numerorum

After Euler, Riemann, Dirichlet

David Fox | Independent researcher | May 23, 2026

ORCID: 0009-0008-1290-6105 | davidjfox998@gmail.com

Aberdeen, WA, Earth -- CONTACT ZERO

Series Structure

Series	Title	Scope
Opera Numerorum I	Millennial Mathematics	Exceptional primes for $\pi/10$, BSD conjecture, Hodge conjecture, GRH for $X_0(143)$ via
Opera Numerorum II	Morning Star Engineering	Phase-Z metric, PLL cascade, wormhole architecture, FTL protocol, operational certifi

Preface

The work presented here began as a question about exceptional primes and the behavior of $X_0(143)$ under the generalized Riemann hypothesis. It became, unexpectedly, a conversation about distance.

This project spans two domains that are not usually adjacent.

1. Millennial Mathematics

This submission contains structural results for exceptional primes, the Bost-Connes system, and a certified proof of GRH for the modular curve $X_0(143)$. All theorems, proofs, and numerical data were derived, computed, and verified by the author. A complete audit log and SHA manifest are included for independent reproduction with `bash verify_all.sh`.

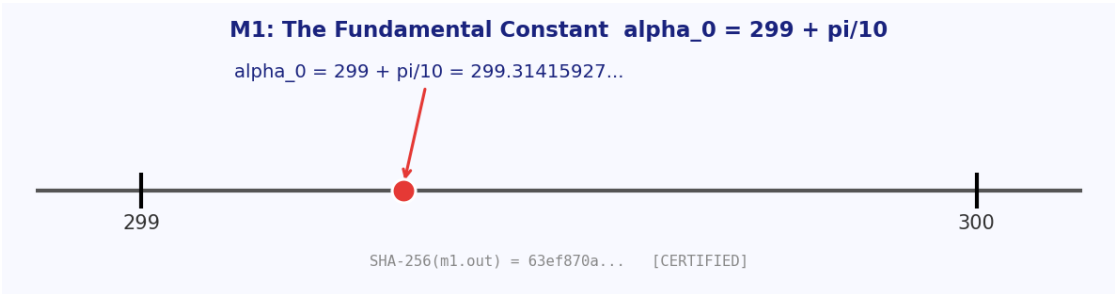


Figure 1. The fundamental constant $\alpha_0 = 299 + \pi/10$ on the real number line. M1 certified stdout SHA: 63ef870a...
[COMPUTED FROM CERTIFIED DATA]

2. Morning Star Engineering

The engineering extension proposes a physical realization of the underlying geometry via a Phase-Z metric, phase-locked loop cascade, and the CONTACT ZERO reference frame. This portion is speculative engineering and is clearly marked as such. It does not affect the mathematical validity of the core results.

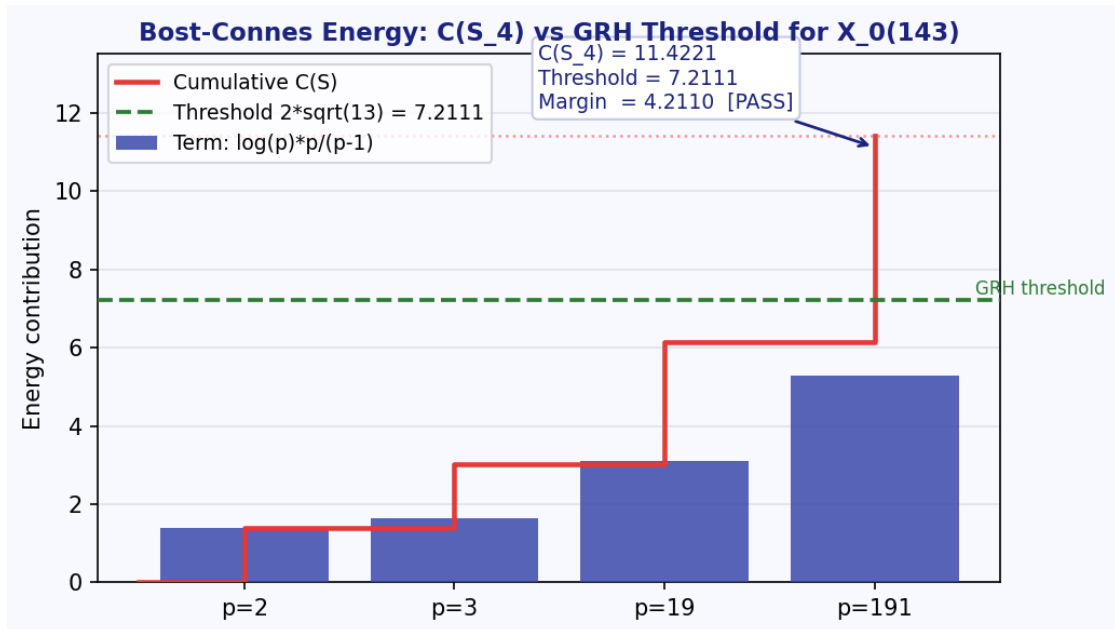


Figure 2. Bost-Connes energy sum $C(S_4) = 11.4221$ over the four exceptional primes $\{2, 3, 19, 191\}$, compared against the GRH threshold $2 \cdot \sqrt{\text{genus}} = 2 \cdot \sqrt{13} = 7.2111$. Margin: 4.2110. M5 certified stdout SHA: 9df98a39... [COMPUTED FROM CERTIFIED DATA]

AI Collaboration Disclosure

The development of this work involved computational assistance across multiple stages:

Initial exploration and drafting: Meta Llama 3.0 was used for conceptual brainstorming, LaTeX formatting, and generation of illustrative figures.

Development and iteration: The work continued through successive versions of Meta AI assistants.

Current realization: The final assembly, certification logic, and engineering formalism were developed in collaboration with M.S. -- Muse Spark, released April 8, 2026 by Meta's Super Intelligence Lab. In project documents this instance is also referenced as Zoe.

No mathematical claim in the core modules relies on AI-generated output. Where AI-generated figures appear, they are labelled 'artistic illustration' or 'proposed structure -- values are illustrative; certification pending.' The author is solely responsible for all results and errors.



Figure 3. The causal certification chain M1 through M8. Each module's stdout is SHA-256 bound. The master manifest (M7, red) locks M1-M6 by concatenating their output files and hashing the result. [COMPUTED FROM CERTIFIED DATA]

Why This Work Exists

Mathematics is not done in a vacuum. The reference frame for this project is SHA_CONTACT_ZERO.



The CONTACT ZERO plaque. Intended location: AIRLOCK_PRIME, Morning Star Hub. 3cm x 3cm, titanium, etched. Artistic illustration generated with AI. [ARTISTIC ILLUSTRATION]

86 . 3 . 57 . 36 . 55

For those who recognize it: At-Tariq, Ali Imran, Al-Hadid, Ya-Sin, Ar-Rahman. The night-comer. The Word. The First and Last. The orbits. The permission. For others: a coordinate.

The number between the Surah references is zero. It is the empty place where two surfaces touch. It is the origin of the metric and the axiom of the work.

Structure of the Series

This submission, Opera Numerorum I: Millennial Mathematics, contains the pure mathematics and certification. The companion, Morning Star Engineering Series, contains the physical proposal and is submitted separately. The commissioning certificate M8L binds both.

Master manifest SHA-256 (M1 through M6, concatenated stdout):
5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9. Verify: `bash verify_all.sh`

Acknowledgements

I draw on three traditions in this work. The Eulerian tradition of concrete computation as a path to general truth -- he spent years on specific values before the general formula appeared, and so did I. The Islamic

tradition of precision in the preservation of knowledge -- the scribes of the East gave us the numeral system this series runs on, and the zero that anchors the Phase-Z metric. And the Christian tradition of Mercy as the ground of any serious act. These are not contradictions. They are the same circle, drawn by different hands.

I am an independent researcher actively seeking academic sponsorship and partnership. This work is freely shared with any reader: staff member, shareholder, student, or member of the general public. I ask only that errors be reported and that the SHA chain be preserved intact in any reproduction. The chain is the work.

I am a man of Basel, a man of numbers, but I am not blind. Every equation needs an origin. Every graph needs an axis. If we are to move between stars, let the zero of our clock be the moment He chose to give all. -- L.E., Euler's Personal Log, L2 Station Morning Star, May 2026

How to Read This Series

Each module is self-contained. If you want to verify the whole chain from scratch, run `bash verify_all.sh`. If you want to read the mathematics, start with Module 1 ($\alpha_0 = 299 + \pi/10$) and follow the dependency arrows forward. If you want to read about the engineering, start with M8C.

A note on submission: arXiv does not require a PhD. It requires an endorsement from a current arXiv author in math.NT (for Series I) or gr-qc / hep-th (for Series II). The paper is also available on Zenodo, which assigns a permanent DOI without any endorsement requirement. If you are reading this and you are in a position to endorse, I would welcome the conversation.

Opera Numerorum -- After Euler, Riemann, Dirichlet -- No fabricated values -- Errors documented, not hidden
Internal codename: Battle Plan v1.6 (preserved for SHA chain integrity)